PREDICTING CERVICAL CANCER

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

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COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I/We, hereby declare that the project report entitled "**PREDICTING CERVICAL CANCER**" is an original work done in the Department of Computer Science and Engineering, GITAM School of Technology, GITAM (Deemed to be University) submitted in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering. The work has not been submitted to any other college or University for the award of any degree or diploma.

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CERTIFICATE

This is to certify that the project report entitled **"PREDICTING CERVICAL CANCER"** is a bonafide record of work carried out by **M.N.S. Shanmukhi (121910306003), Vinoothna More (121910306023), K. Durga Sravya (121910306056), Vineel Mulakala (121910306046)** students submitted in partial fulfillment of requirement for the award of degree of Bachelors of Technology in Computer Science and Engineering.

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1. ABSTRACT:

An increasing number of people and businesses are using machine learning (ML) and deep learning (DL) to analyze vast volumes of data and provide useful insights. In medical practice, utilizing ML-based techniques to predict the early stages of catastrophic illnesses like cancer, renal failure, and heart attacks is becoming increasingly widespread.

Cervical cancer is one of the most frequent illnesses among women, and early identification may be an effective method of prevention. It is still one of the most frequent gynecologic cancers worldwide. Early screening is the most effective technique for lowering the worldwide burden of cervical cancer since it is a highly preventable illness. Unfortunately, disadvantaged patient groups in poor nations cannot afford to be tested on a frequent basis due to a lack of information, access to medical institutions, and the high cost of operations. Healthcare providers may use machine learning algorithms to increase the accuracy of cervical cancer diagnosis, forecast which people are at risk of acquiring the illness, and build individualized treatment strategies for each patient.

Machine learning models can be trained to analyze the patient's age, sexual history, smoking habits, family history, and other health indicators to predict the likelihood of cervical cancer. They can also be used to analyze medical images and pathology data to identify early warning signs of cancer.

In this study, a unique ensemble method for estimating the risk of cervical cancer is presented. By implementing a voting approach, this method solves the problems with earlier research on cervical cancer. A data rectification technique is suggested in order to enhance prediction performance. In order to evaluate the suggested procedure, many measures are made. The results imply that the voting method may accurately estimate a person's risk of getting cervical cancer.

2. INTRODUCTION:

The fourth most common malignant tumor in women in 2018 was cervical cancer, which accounted for 6.6% of all female malignancies diagnosed. The progression of normal cells from a precancerous lesion to invasive malignant cervical cancer, which includes a protracted and reversible precancerous lesion stage, can take years or decades. Yet, it is well known that early detection and treatment of precancerous lesions can stop the progression of the disease in 90% of individuals with cervical cancer.

Significant variations in the risk of HPV cancerization at various ages have been found in several studies. Due to socioeconomic position and educational level, the majority of women are uninformed about the causes, risk factors, prevention, and treatment of cervical cancer, despite the fact that it is a disease that is largely preventable. The incidence of cervical cancer is still lower in developed nations than in developing nations. In actuality, 95% of cervical cancer fatalities occur in low-income nations. HPV infection, which is transferred through intercourse, is another significant cause. As a result, a higher risk of cervical cancer has been linked to factors such as the age of the first sexual encounter, the quantity of sexual partners, the number of pregnancies, and the usage of contraception. If these modifiable risk factors are effectively managed and the natural prognosis of cervical cancer for those with precancerous cervical lesions is modified, the incidence of cervical cancerization can be avoided.

The majority of cervical cancer diagnoses are based on the colposcopy biopsy results. Yet, many women are reluctant to have colposcopic procedures performed, and misusing colposcopy would be a waste of money on healthcare. This study's objective was to examine cervical cancer risk factors and build an ensemble prediction model using them.

3. LITERATURE REVIEW:

1. Jiayi Lu, Enmin Song, Ahmed Ghoneim, Mubarak Alrashoud, "Machine learning for assisting cervical cancer diagnosis: An ensemble approach". January 2020

In this work, we looked at a variety of cervical cancer prevention strategies and suggested a brand-new, effective auxiliary model for cervical cancer prediction utilizing a gene sequence module. Also, the proposed approach is more adaptable and useful when compared to previous methods. The findings indicate that machine learning in medical research has virtually endless potential, despite the suggested approach's limitations due to experimental support. Our next research willconcentrate on more unique data, such as colposcopy pictures.

2. ***2. M. Chen, X. Shi, Y. Zhang, and colleagues, Deep features learning for medical image analysis using a convolutional autoencoder neural network, IEEE Trans. Big Data, vol (2017).***

Computed tomography (CT) is currently routinely utilized to aid in illness diagnosis. Computer assisted diagnosis (CAD) based on artificial intelligence (AI) has lately demonstrated its usefulness in intelligent healthcare. Due to privacy and security concerns, establishing an acceptable labeled dataset for CT analytic assistance is a significant task. As a consequence, this research offers a convolutional autoencoder deep learning framework to enable unsupervised image feature learning for lung nodules using unlabeled data, which only requires a small amount of labeled data for successful feature learning. Extensive trials indicate that the suggested strategy outperforms alternative approaches in successfully solving the inherent labor-intensivechallenge during artificial picture labeling. It also shows how the recommended convolutional autoencoder approach may be expanded to measure how similar lung nodule images are. The characteristics collected by unsupervised learning, in particular, are relevant in other similar circumstances.

3. A. Ghoneim, G. Muhammad, M.S. Hossain, Cervical cancer classification using convolutional neural networks and extreme learning machines, Future Gener. Comput. Syst. 102 (2020) 643–649.

For cervical cancer detection and classification, a CNN-based method was proposed. The ELM-based classifier or the AE-based classifier was utilized when the CNN model was added to the system. Both shallow and deep CNN models were looked into. The suggested system

with the ELM-based classifier successfully completed the 2-class problem with 99.7% accuracy using the Herlev database and the 7-class problem with 97.2% accuracy. Their accuracy levels exceeded any previously reported accuracy levels.

4. H.L. Fatlawi, Enhanced classification model for cervical cancer dataset based on cost sensitive classifier, Int. J. Comput. Tech. 4 (4) (2017) 1–8.

Unreliable results may emerge from enhancing a classification model without taking the true cost of each mistake event into account. A decision tree classifier with a cost matrix with various cost values is the foundation of the suggested model. In circumstances where medical testing indicate that a patient is infected but they are labeled as not being diseased, there is a larger cost for inaccuracy. In both binary class and multiclass classification, the recommended approach produces more accurate results. Its TP rate is (0.429), while the TP rate for the standard binary class decision tree job is (0.160).

4. PROBLEM IDENTIFICATION AND OBJECTIVES:

The cells of the cervix, which is the lowest portion of the uterus that links to the vagina, are where cervical cancer begins. Regular screening tests and follow-up care, which can detect precancerous cells before they develop into cancer, can often prevent cervical cancer.

The problem of cervical cancer prediction is to create a model that can accurately predict a woman's risk of developing cervical cancer based on various risk factors such as age, sexual behavior, smoking habits, family history, and previous cervical cancer diagnosis. This prediction model's goal is to assist healthcare providers in identifying women who may be at higher risk for cervical cancer and recommend appropriate preventive measures such as regular screenings, HPV vaccination, and lifestyle changes.

Machine learning algorithms and statistical methods can be used to develop the prediction model, which can analyze large amounts of data and identify patterns and trends. The model can be trained on a dataset of women with known cervical cancer outcomes and risk factors, and then validated on a separate dataset to determine its accuracy and generalizability. Finally, the cervical cancer prediction model has the potential to be a valuable tool in the prevention and early detection of cervical cancer, resulting in better health outcomes and lower healthcare costs.

4.1 Existing Model:

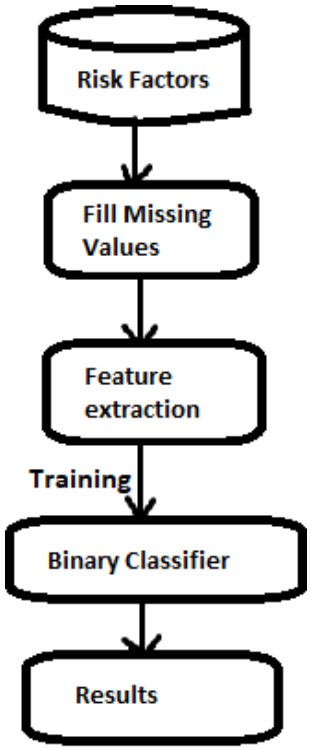
In recent years, interest in machine learning-related medical research has increased. Because of this, more and more scientific research teams are keen to publish their datasets online. A dataset including risk factors for cervical cancer was made available through the UCI Machine Learning Repository in 2017. Afterwards, additional researchers conducted studies utilizing this data to forecast cervical cancer.

Apart from the original data, the initial analysis on the dataset may be traced back to a cost-sensitive classifier, the accuracy of which on the cervical data only just above the basic level. To predict the risk of cervical cancer, two improved support vector machine (SVM) techniques, SVM-recursive feature elimination and SVM-principal component analysis (SVM-PCA), were developed. Feature selection procedures were given more consideration in subsequent investigations. Recently, an unique feature selection method called the Firefly Algorithm and Random Forest Classifier was suggested. Similar to this, Sherif et al. used the synthetic minority oversampling method to minimize the number of features based on Random Forest classification (SMOTE). A stacked autoencoder with a softmax layer has achieved an accuracy of 97.25% in the cervical cancer dataset.

Cancer prediction involves two steps: training and testing. Missing values are filled in and features are obtained and processed into the proper forms during the training phase. As input, the classifier takes the data and adjusts its weights to find convergence. During the testing step, the performance of the classifier is evaluated. However, these results are based only on a single, highly skewed cervical cancer dataset. Also, these tests' measurements are rather homogeneous.

Disadvantages:

- Low Accuracy
- Less no.of risk factors are considered
- No colposcopy images used



Existing Model

4.2 Proposed Model:

Due to patients' reluctance to answer certain questions out of concern for their privacy or because the answers to some questions, including those on familial-hereditary disorders, may not be known, missing data is unavoidable throughout the data collecting phase. Directly removing the missing data is a simple method that produces a reduced dataset. Nowadays, some studies estimate the missing values using random forest, but this method is constrained by the values of other characteristics. As a result, we chose a filling technique with a corrective mechanism that is quite straightforward. The traits are connected, according the interpretation of the data. For instance, the age of the first pregnancy must be greater than or equal to the age of the first sexual experience, and the number of pregnancies cannot be less than the number of births. We carefully considered logic when creating the rectification procedure. The correction procedure is initially applied to the current numbers since mistakes might happen when sorting the data. So, using the appropriate data limitation mode method, the missing values

are assigned with values. Moreover, certain qualities are essential to the prediction task and cannot be eliminated, while some values are yet unknown. For instance, while the findings of HPV tests are essential for the prognosis of cervical cancer, some patients have not yet completed the necessary testing. In this case, this property receives an additional value that represents the unknown.

In the medical field, determining a person's illness accurately is more important than determining their wellness. As a result, accuracy is not the sole factor to consider when evaluating an algorithm. Four criteria are used in this study to examine different algorithms for predicting cervical cancer: accuracy, recall, precision, and F1 score. Because of its straightforward meaning, accuracy is commonly used. Recall and precision are two widely used measures for 2- classification tasks with imbalanced datasets. Recall is the frequency with which positive samples are found. The rate at which samples are accurately labeled as positive is known as precision. The F1 score is used to keep two indications in balance when they conflict withone another. To summarize the results of the prediction, TP, TN, FP, and FN are combined to form a confusion matrix. An ill person is projected to be a diseased person, or TP. TN stands for genuine negative, meaning that the healthy individual is appropriately identified as such. False positive (FP) is the abbreviation for when a healthy individual is mistakenly categorized as having an illness. False negative, or FN, refers to an unwell individual who appears to be healthy. It is clear that FN is the most significant element and should be kept as low as feasible. Another influencing element that should be as significant as feasible is TP.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

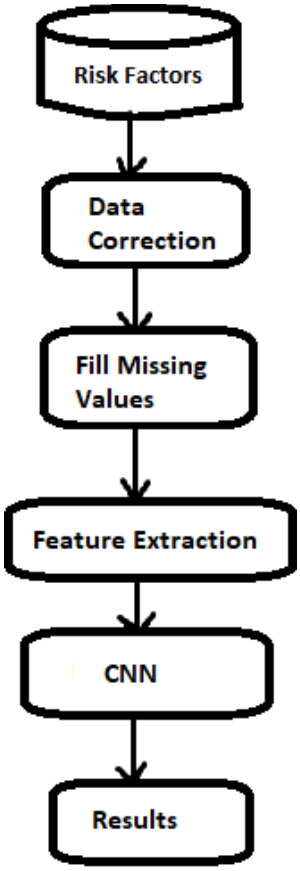
$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$F1score = \frac{2 \times Recall \times Precision}{Recall + Precision}$$

Advantages:

- Increased Accuracy
- More no.of risk factors are considered
- Colposcopy images are used



Proposed Model

5. SYSTEM METHODOLOGY:

5.1 System Requirements Specification

The system requirements for this project include:

- A computer with a GPU for training the machine learning model
- A dataset of cervical images for training and testing
- Image processing and machine learning software

5.2 Software and Hardware Requirements:

The software and hardware requirements for this project include:

- Python programming language

- TensorFlow and Keras libraries for machine learning
- OpenCV library for image processing
- NVIDIA GPU for training the machine-learning model

5.3 System Methodology:

The methodology for this project involves the following steps:

Defining the problem: The cells of the cervix, which is the lowest portion of the uterus that links to the vagina, are where cervical cancer begins. Regular screening tests and follow-up care, which can detect precancerous cells before they develop into cancer, can often prevent cervical cancer.

The problem of cervical cancer prediction is to create a model that can accurately predict a woman's risk of developing cervical cancer based on various risk factors such as age, sexual behavior, smoking habits, family history, and previous cervical cancer diagnosis. This prediction model's goal is to assist healthcare providers in identifying women who may be at higher risk for cervical cancer and recommend appropriate preventive measures such as regular screenings, HPV vaccination, and lifestyle changes.

The prediction model may be created using statistical techniques and machine learning algorithms, and it can examine enormous volumes of data to find patterns and trends. The model can be trained on a dataset of women with known cervical cancer outcomes and risk factors, and then validated on a separate dataset to determine its accuracy and generalizability. Last but not least, the cervical cancer prediction model has the potential to be a useful tool for cervical cancer prevention and early identification, leading to improved health outcomes and lower healthcare expenditures.

Data preprocessing: It is a critical step in any data analysis or machine learning project, including cervical cancer research. Data preprocessing in the context of cervical cancer entails cleaning and transforming raw data obtained from various sources to make it suitable for analysis and modeling.

The steps involved in data preprocessing for cervical cancer prediction:

1. **Data collection:** The first step is to gather information from various sources, such as medical records, surveys, and clinical trials. Patient demographics, medical history, test results, and treatment outcomes should all be included in the data.
2. **Data cleaning:** Missing values, outliers, or errors in the collected data may affect the accuracy of the analysis. As a result, it is critical to clean the data by removing or replacing missing values, correcting errors, and dealing with outliers.
3. **Data transformation:** The collected data may be in various formats or scales, making comparison and analysis difficult. To make the data consistent and meaningful, it is necessary to transform it through scaling, normalization, or encoding categorical variables.
4. **Feature selection:** Many features or variables in the collected data may be irrelevant or redundant, increasing the complexity of the analysis or modeling. As a result, it is critical to choose the relevant features that are most informative and useful for the analysis.
5. **Data splitting:** To evaluate the performance of the models and prevent overfitting, it is crucial to divide the data into training, validation, and test sets once it has been preprocessed.

Overall, data preprocessing is a crucial step in cervical cancer research that helps to ensure the accuracy and validity of the analysis or modeling results.

Exploratory data analysis: Understanding and displaying the patterns and correlations revealed in the cervical cancer dataset requires the use of exploratory data analysis (EDA). EDA aids in locating any outliers, correlations, and other data trends that might provide details about the cervical cancer's underlying causes.

The techniques used in EDA for cervical cancer prediction:

1. **Summary statistics:** The first step in EDA is to compute summary statistics for each variable in the dataset, such as mean, median, standard deviation, and range. This aids in determining the distribution and spread of the data as well as identifying any potential outliers.
2. **Visualization:** In cervical cancer research, a visualization is a critical tool for EDA. Histograms, box plots, scatter plots, and heatmaps are examples of common visualizations. These visualizations aid in the identification of patterns, correlations, and outliers in data.
3. **Correlation analysis:** Correlation analysis is useful for determining the strength and direction of the relationship between two or more variables in a dataset. Correlation analysis in cervical cancer research can aid in the identification of potential risk factors or protective factors for cervical cancer.
4. **Dimensionality reduction:** By highlighting the most crucial characteristics or variables, dimensionality reduction techniques like principal component analysis (PCA) and t-distributed stochastic neighbor embedding (t-SNE) can help reduce dataset complexity.
5. **Cluster analysis:** Cluster analysis aids in the identification of groups or clusters within data based on the similarity or dissimilarity of the features. Cluster analysis in cervical cancer research can help identify subgroups of patients with similar risk factors or treatment outcomes.

Overall, EDA is a critical stage in the investigation of cervical cancer that aids in the discovery of potential patterns, correlations, and trends in the data. EDA can aid in the creation of more precise and efficient models for prediction and diagnosis and offer insightful information about the underlying causes of cervical cancer.

Data Visualization: It is an effective tool for investigating and communicating complex patterns and relationships in the cervical cancer dataset. Visualization can aid in the identification of trends, outliers, and correlations in data, as well as in communicating insights to a wider audience.

The common techniques used in data visualization for cervical cancer research:

1. **Histograms:** Histograms are a popular visualization tool used to investigate the distribution

of a single variable in a dataset. Histograms can aid in the identification of the distribution of age, tumor size, and other key variables in cervical cancer research.

2. **Scatter plots:** Scatter plots are an effective visual tool for investigating the relationship between two variables in a dataset. Scatter plots can aid in the identification of the relationship between age and tumor size, as well as other potential risk factors, in cervical cancer research.

3. **Box plots:** Box plots are an excellent tool for visualizing the distribution of a variable across multiple groups or categories. Box plots can be used in cervical cancer research to identify differences in tumor size or other variables between different stages of cancer.

4. **Heatmaps:** Heatmaps are a type of visualization tool that can assist in identifying patterns or correlations between multiple variables in a dataset. Heatmaps can aid in the identification of potential risk factors or protective factors in cervical cancer research.

5. **Network diagrams:** Network diagrams are a type of visualization that can be used to investigate complex relationships between multiple variables in a dataset. Network diagrams can aid in the identification of the relationship between various risk factors and treatment outcomes in cervical cancer research.

Overall, data visualization is an essential tool for exploring and communicating patterns and relationships within the cervical cancer dataset. Visualization can help to identify potential risk factors, correlations, and trends in the data, and guide the development of more accurate and effective models for prediction and diagnosis.

Balancing the dataset: In cervical cancer research, balancing the dataset entails ensuring that the number of observations or samples in each class or category is roughly equal. This is significant because biased or inaccurate model predictions and poor generalization performance can result from imbalanced datasets.

Techniques that can be used to balance the dataset in cervical cancer research:

1. **Undersampling:** To match the number of observations in the minority class, a subset of observations from the majority class are chosen at random and subsampled. This technique may help to reduce the imbalance in the dataset, but it may also result in data loss.

2. **Oversampling:** Replicating minority class observations to match the number of observations in the majority class is the process of oversampling. This technique may help to increase minority class representation, but it may also result in overfitting.
3. **Synthetic data generation:** Synthetic data generation entails generating new observations that are similar to existing ones in order to create new samples in the minority class. This technique can help to increase minority class representation while avoiding overfitting.
4. **Cost-sensitive learning:** Cost-sensitive learning entails varying the weights or costs assigned to each class based on their relative frequency or importance. This technique can help to heavily penalize errors in the minority class and improve the model's performance on the minority class.
5. **Ensemble methods:** To improve overall performance, ensemble methods combine multiple models trained on different subsets of the dataset. This method can aid in enhancing the model's ability to generalize to the minority class.

Overall, to achieve accurate and impartial model predictions, balancing the dataset is a crucial step in cervical cancer research. Depending on the particular dataset and research issue, many strategies may be employed, and it is crucial to assess how well the model performs for both minority and majority groups.

Building the model: Creating a cervical cancer model entails creating a predictive algorithm that can accurately classify patients as having or not having cervical cancer based on relevant clinical or demographic features.

Here are some common techniques for developing a cervical cancer model

1. **Feature selection:** The process of selecting the most relevant clinical or demographic features that are predictive of cervical cancer is known as feature selection. Age, HPV status, tumour size, and stage are all common features in cervical cancer models.
2. **Model selection:** Model selection entails selecting the best machine learning algorithm for the dataset and research question. Logistic regression, decision trees, random forests, support vector machines, and neural networks are often used methods in the study of cervical cancer.
3. **Cross-validation:** To evaluate the model's performance on new data, cross-validation divides the dataset into training and validation sets. Cross-validation techniques commonly used include

k-fold and leave-one-out cross-validation.

4. **Hyperparameter tuning:** Hyperparameter tuning is the process of fine-tuning the machine learning algorithm's parameters to enhance model performance. In hyper parameter tuning, grid search, random search, and Bayesian optimization are all frequently utilized methods.

5. **Evaluation metrics:** The effectiveness of the model is assessed using assessment metrics. In cervical cancer research, popular assessment criteria include sensitivity, specificity, accuracy, area under the receiver operating characteristic curve (AUC-ROC), and F1 score.

Overall, building a model for cervical cancer involves a combination of feature selection, model selection, cross-validation, hyperparameter tuning, and evaluation metrics to develop an accurate and effective algorithm for predicting cervical cancer. It is important to consider the clinical relevance of the model and the potential impact on patient outcomes.

Testing the model: In cervical cancer prediction, testing the model entails evaluating its performance on a test dataset that was not used during model development. This step is critical to ensuring that the model generalizes well to new, previously unseen data and does not overfit the training dataset.

Here are some common techniques used in testing the model in cervical cancer prediction:

1. **Holdout validation:** Holdout validation involves creating training and testing sets from the dataset at random, with the training set being used to create the model and the testing set being used to evaluate its effectiveness. Because of the randomness of the data split, this technique is simple and quick, but it may result in a high variance estimate of the model's performance.

2. **Cross-validation:** In cross-validation, the dataset is divided into several folds, the model is trained on various subsets, and its performance is assessed on the full dataset. This strategy can produce a more reliable assessment of the model's performance and is less affected by the randomness of the data split.

3. **External validation**: External validation entails testing the model using a dataset that was not used during model construction. This technique provides the most stringent test of the model's generalizability, although it is potentially constrained by the availability of external

datasets.

4. **Evaluation metrics:** To measure the model's performance on the test dataset, evaluation metrics like sensitivity, specificity, accuracy, AUC-ROC, and F1 score are utilized. It is critical to select evaluation measures that are relevant to the study issue and clinical situation.

Overall, testing the model in cervical cancer prediction is a critical step in model development and validation. It is important to choose appropriate techniques and evaluation metrics to ensure that the model generalizes well to new data and is clinically relevant.

Deployment: Making a cervical cancer prediction model available for use in clinical practice or research is what deployment entails. The following are some frequent steps in the deployment process:

1. **Model optimization**: It is critical to optimize the model before deploying it to guarantee that it operates effectively and properly. This could include modifying hyperparameters, improving feature selection, or using overfitting-reduction approaches.
2. **Model integration:** The model must be incorporated into the research system or clinical process it will be used in. Integrating the model with electronic health records, imaging systems, orother relevant data sources may be required.
3. **User interface:** The model should have an easy-to-use interface that is simple to explore and understand. This could entail creating a web-based application, a mobile app, or another software interface that allows users to enter data and view predictions.
4. **Validation and evaluation:** After deployment, the model should be tested and reviewed to ensure that it continues to perform well in real-world scenarios. This could include tracking the accuracy of the forecasts, measuring the influence on patient outcomes, or asking for user input.
5. **Maintenance and updates:** To guarantee that the model continues to perform well over time, it should be maintained and updated as needed. This could include adding additional data to the model, upgrading the user interface, or introducing new features or methodologies.

Overall, the deployment of a cervical cancer prediction model requires careful planning and

execution to ensure that the model is effective, efficient, and usable in clinical practice or research. It is important to work closely with clinicians, researchers, and other stakeholders to ensure that the model is clinically relevant and meets the needs of the intended users.

6. OVERVIEW OF TECHNOLOGIES:

Pandas - Pandas, an open-source Python tool, is used to analyze and manipulate data. It provides data structures and procedures for interacting with numerical tables and time series data. The two primary data structures provided by Pandas are Series and DataFrame. A series is a one-dimensional labeled array that may hold any type of data, whereas a data frame is a two-dimensional labeled data structure with columns of potentially diverse types. Moreover, Pandas provides capabilities for working with missing data, handling time series data, merging and combining datasets, and much more. Data preparation, cleaning, analysis, and visualization using tools like Matplotlib or Seaborn are all common uses for Pandas.

Numpy - NumPy is a well-known open-source Python tool for data processing and scientific computing (Numerical Python). Massive, multidimensional arrays and matrices are supported, and they may be used for a broad variety of intricate mathematical operations. A key library for Python's scientific computing, NumPy serves as the foundation for several other libraries, including Pandas and Scikit-learn. Due to its strong array manipulationcapabilities and effective numerical calculation features, NumPy is frequently used in data analysis, scientific computing, and machine learning activities.

Matplotlib - Matplotlib is a popular open-source Python toolkit used for data visualization. It provides a selection of customizable graphs and charts to simply and effectively represent data. In 2003, John D. Hunter developed Matplotlib, which has since developed into one of the most widely used data visualization tools in the Python environment. The plotting functions in Matplotlib allow you to create a wide range of graphs, including line plots, scatter plots, bar plots, histograms, and more. These graphs can have their colors, labels, fonts, and other design features changed. Due to its capability to produce

stunning representations of large, complicated data sets, Matplotlib is frequently used in scientific research, data analysis, and machine learning activities.

Seaborn - An open-source Python module called Seaborn is used for statistical graphics and data visualization. It is based on Matplotlib and offers a more advanced user interface for producing attractive and instructive statistical visualizations. With the help of the scatter plots, line plots, bar plots, and heatmaps, among other statistical plots that Seaborn provides, you can see how variables are correlated. Moreover, it offers sophisticated visualization methods that may be used to investigate complicated data sets, including violin plots, swarm plots, and factorplots. Due to its capability to produce stunning and useful representations of large data sets, Seaborn is frequently used in data analysis, scientific research, and machine learning activities.

Xgboost - A well-liked open-source machine learning toolkit called XGBoost (eXtreme Gradient Boosting) is utilized for supervised learning issues including classification and regression applications. The gradient boosting algorithm, an ensemble learning technique that combines many weak models to produce a stronger model, is the foundation of XGBoost. Gradient boosting's main principle is to iteratively train new models on the residual errors of older models in order to decrease the ensemble model's overall error. Due to its capacity to develop precise and effective models for a variety of supervised learning tasks, XGBoost is frequently utilized in data science contests like Kaggle.

Sklearn - A well-liked open-source machine learning package for Python is called Scikit-learn, sometimes referred to as sklearn. In addition to tools for data preparation and model selection, it offers a wide choice of algorithms for tasks including classification, regression, clustering, and dimensionality reduction. Since its original release in 2007, Scikit-learn has grown to rank among the most popular machine learning libraries in the Python community. It is constructed on top of several well-known libraries for scientific computing, including NumPy, SciPy, and Matplotlib. Due to its simplicity of use, rich documentation, and variety of machine learning methods and tools, Scikit-learn is extensively used in data science, academia, and industry.

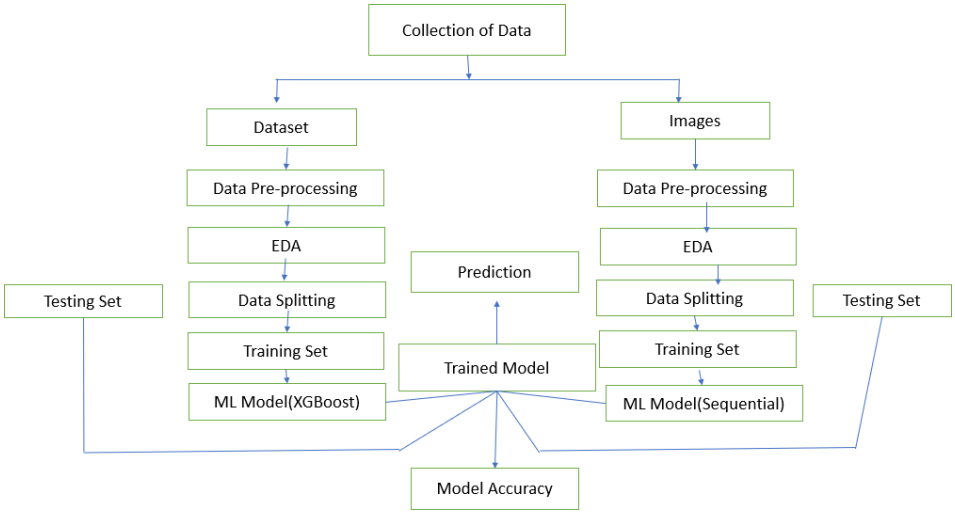
Tensorflow - The open-source machine learning software package known as TensorFlow was developed in 2015 by the Google Brain Team. It is made to offer a user-friendly and adaptable environment forcreating and honing machine learning models. TensorFlow is founded on the idea of a

computational network, in which data flows via nodes that represent mathematical processes. This enables effective parallel processing and computation optimization. TensorFlow is widely used in a variety of applications, including speech recognition, natural language processing, and computer vision. It has also been used for a variety of tasks in research and business, including speech and picture recognition, language translation, and anomaly detection.

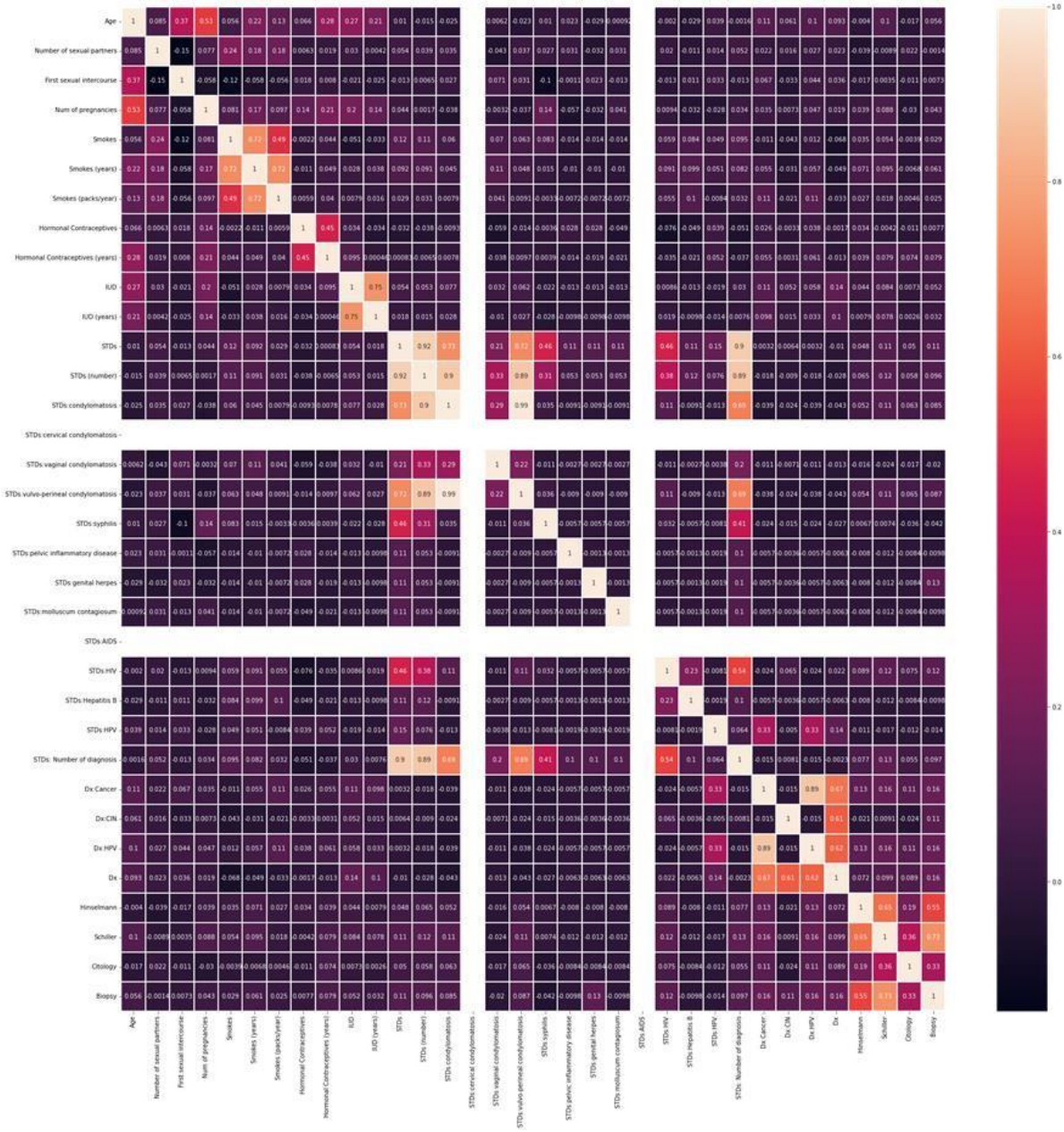
Pickle - Python's Pickle library is used to serialize and deserialize Python objects. Users may do this to transform Python objects into a format that can be saved in a file or sent over a network, and then those objects can be reconverted back into Python objects. De-serialization is the process of transforming a serialized object back into a Python object. Serialization is the act of changing a Python object into a format that can be stored. Pickle is a strong and practical tool for storing and moving data across various Python applications or between Python programs themselves.

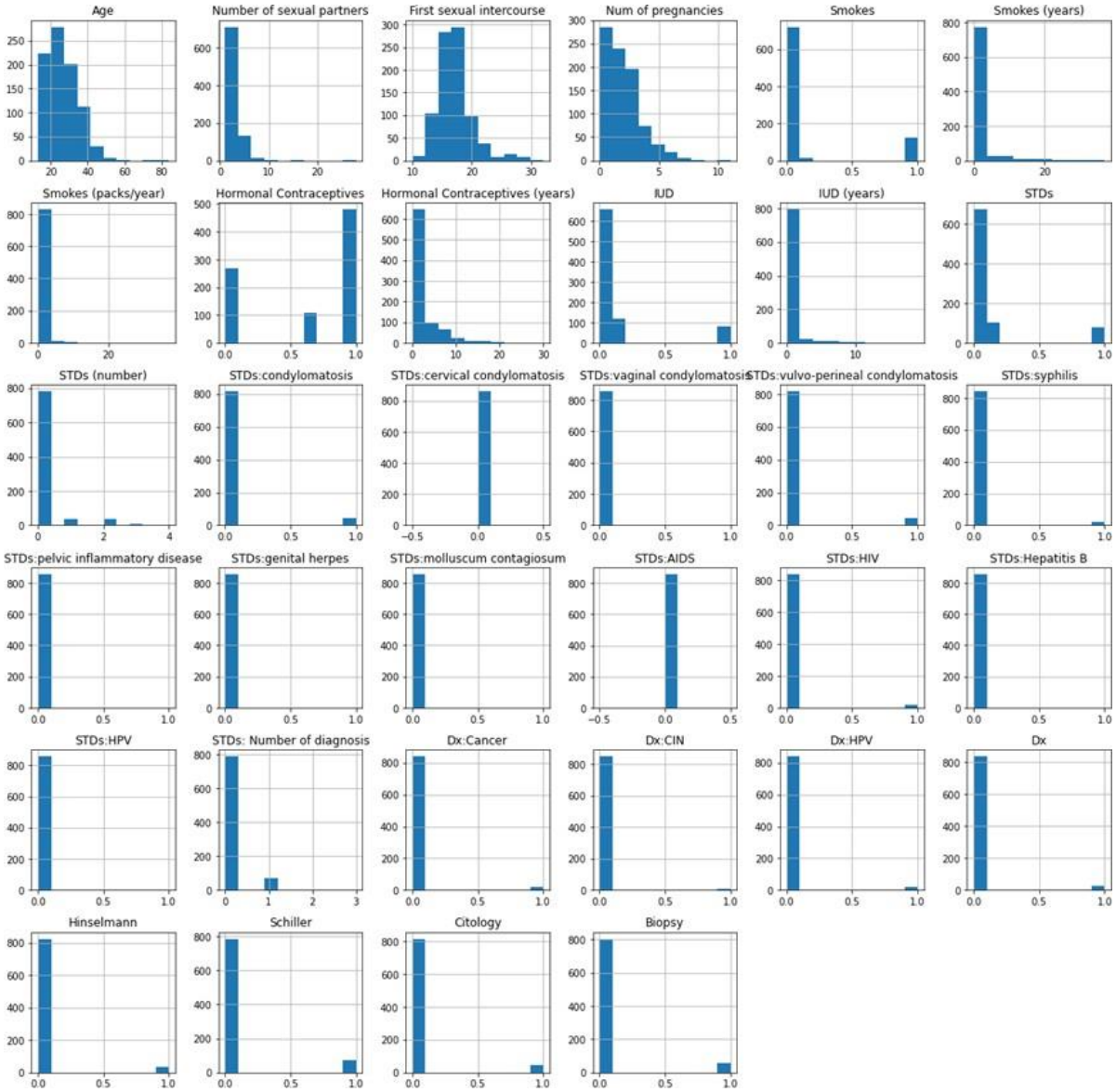
Keras - The TensorFlow, Theano, and CNTK machine learning frameworks serve as the foundation for the high-level, open-source Keras neural network library, which was developed in Python. It provides a simple and clear interface for building and improving deep learning models. Deep neural network testing may be done quickly thanks to Keras, which enables users to quicklyprototype and improve their models. It includes a wide range of pre-built layers that may be used to build complex models, including as dense, recurrent, and convolutional layers. Because of its simplicity, adaptability, and applicability, Keras is becoming more and more well-liked in the deep learning field.

7. IMPLEMENTATION:



Heatmap:


Age
Number of sexual partners
First sexual intercourse
Num of pregnancies
Smokes
Smokes (years)
Smokes (packs/year)
Hormonal Contraceptives
Hormonal Contraceptives (years)
IUD
IUD (years)
STDs
STDs (number)
STDs condylomatosis
STDs cervical condylomatosis
STDs vaginal condylomatosis
STDs vulvo-perineal condylomatosis
STDs syphilis
STDs pelvic inflammatory disease
STDs genital herpes
STDs molluscum contagiosum
STDs AIDS
STDs HIV
STDs Hepatitis B
STDs HPV
STDs: Number of diagnosis
Dx Cancer
Dx CIN
Dx HPV
Dx
Hinselmann
Schiller
Citology
Biopsy


Age
Number of sexual partners
First sexual intercourse
Num of pregnancies
Smokes
Smokes (years)
Smokes (packs/year)
Hormonal Contraceptives
Hormonal Contraceptives (years)
IUD
IUD (years)
STDs
STDs (number)
STDs:condylomatosis
STDs:cervical condylomatosis
STDs:vaginal condylomatosis
STDs:vulvo-perineal condylomatosis
STDs:syphilis
STDs:pelvic inflammatory disease
STDs:genital herpes
STDs:molluscum contagiosum
STDs:AIDS
STDs:HIV
STDs:Hepatitis B
STDs:HPV
STDs: Number of diagnosis
Dx:Cancer
Dx:CIN
Dx:HPV
Dx
Hinselmann
Schiller
Citology
Biopsy

7.1 CODING:

In [2]:
```
#read dataset
cancer_df = pd.read_csv(r"C:\Users\vinoo\Downloads\cervical_cancer_dataset.csv")

pd.set_option('display.max_columns',None)
#print dataset with head(5)
cancer_df.head(5)
```

Out[2]:

	Age	Number of sexual partners	First sexual intercourse	Num of pregnancies	Smokes	Smokes (years)	Smokes (packs/year)	Hormonal Contraceptives	Hormonal Contraceptives (years)	IUD	IUD (years)	STDs	STDs (number)	STDs:condylomatosis
0	18	4.0	15.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	15	1.0	14.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	34	1.0	?	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	52	5.0	16.0	4.0	1.0	37.0	37.0	1.0	3.0	0.0	0.0	0.0	0.0	0.0
4	46	3.0	21.0	4.0	0.0	0.0	0.0	1.0	15.0	0.0	0.0	0.0	0.0	0.0

In [3]:
```
#shape of dataset
cancer_df.shape
```

Out[3]: (858, 36)

In [27]:
```
import xgboost as xgb

model = xgb.XGBClassifier(learning_rate = 0.1, max_depth = 50, n_estimators = 100)

model.fit(X_train, y_train)
```

Out[27]:
```
XGBClassifier(base_score=0.5, booster='gbtree', callbacks=None,
              colsample_bylevel=1, colsample_bynode=1, colsample_bytree=1,
              early_stopping_rounds=None, enable_categorical=False,
              eval_metric=None, gamma=0, gpu_id=-1, grow_policy='depthwise',
              importance_type=None, interaction_constraints='',
              learning_rate=0.1, max_bin=256, max_cat_to_onehot=4,
              max_delta_step=0, max_depth=50, max_leaves=0, min_child_weight=1,
              missing=nan, monotone_constraints='()', n_estimators=100,
              n_jobs=0, num_parallel_tree=1, predictor='auto', random_state=0,
              reg_alpha=0, reg_lambda=1, ...)
```

In [28]:
```
result_train = model.score(X_train, y_train)

result_train
```

Out[28]: 0.9970845481049563

In [29]:
```
result_test = model.score(X_test, y_test)

result_test
```

Out[29]: 0.9186046511627907

In [28]:
```
result_train = model.score(X_train, y_train)

result_train
```

Out[28]: 0.9970845481049563

In [29]:
```
result_test = model.score(X_test, y_test)

result_test
```

Out[29]: 0.9186046511627907

In [30]:
```
y_predict = model.predict(X_test)
```

In [31]:
```
from sklearn.metrics import confusion_matrix, classification_report

print(classification_report(y_test, y_predict))
```

```
              precision    recall  f1-score   support

         0.0       0.93      0.99      0.96        77
         1.0       0.75      0.33      0.46         9

    accuracy                           0.92        86
   macro avg       0.84      0.66      0.71        86
weighted avg       0.91      0.92      0.90        86
```

```
pred_train = np.round(pred_train)
pred_test = np.round(pred_test)

train_acc = accuracy_score(y_train,pred_train)
test_acc = accuracy_score(y_test,pred_test)
train_recall = recall_score(y_train,pred_train)
test_recall = recall_score(y_test,pred_test)
train_precision = precision_score(y_train,pred_train)
test_precision = precision_score(y_test,pred_test)
train_f1 = f1_score(y_train,pred_train)
test_f1 = f1_score(y_test,pred_test)

print('Train accuracy:',train_acc)
print('Test accuracy:',test_acc)
print('Train Recall:',train_recall)
print('Test Recall:',test_recall)
print('Train precision:',train_precision)
print('Test precision:',test_precision)
print('Train f1 score:',train_f1)
print('Test f1 score:',test_f1)
```

```
Train accuracy: 0.995
Test accuracy: 0.9612403100775194
Train Recall: 0.9166666666666666
Test Recall: 0.7368421052631579
Train precision: 1.0
Test precision: 0.7368421052631579
Train f1 score: 0.9565217391304348
Test f1 score: 0.7368421052631579
```

In [4]:
```
root_dir = r'C:\Users\vinoo\Cancer\final';
classes = ["Dyskeratotic","Koilocytotic","Metaplastic","Parabasal","Superficial-Intermediate"]

def GetDatasetSize(path, classes, main = "CROPPED"):
    num_of_image = {}
    for cls in classes:
        # Counting the Number of Files in the Folder
        num_of_image[cls] = len(os.listdir(os.path.join(path, cls, main)));
    return num_of_image;

print(GetDatasetSize(root_dir, classes, "COMPLETE"));
```

```
{'Dyskeratotic': 223, 'Koilocytotic': 238, 'Metaplastic': 271, 'Parabasal': 108, 'Superficial-Intermediate': 126}
```

Split the Dataset such that we have

70% for Train Data

15% for Validation Data

15% for Testing Data

Model Building

In [13]:
```
# Importing Keras for Image Classification
import keras
from keras.layers import Dense,Conv2D, Flatten, MaxPool2D, Dropout
from keras.models import Sequential
# from keras.preprocessing import image
from keras.callbacks import ModelCheckpoint
from keras.models import load_model
```

In [14]:
```
# CNN Model

model = Sequential()
# Convolutional Layer with input shape (64,64,3)
model.add(Conv2D(filters=16, kernel_size= (3,3), activation= 'relu', input_shape=(64,64,3)) )

model.add(Conv2D(filters=32, kernel_size=(3,3), activation='relu' ))
model.add(MaxPool2D(pool_size=(2,2)))


model.add(Conv2D(filters=64, kernel_size=(3,3), activation='relu' ))
model.add(MaxPool2D(pool_size=(2,2)))


model.add(Conv2D(filters=128, kernel_size=(3,3), activation='relu' ))
model.add(MaxPool2D(pool_size=(2,2)))
```

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 conv2d (Conv2D)             (None, 62, 62, 16)        448

 conv2d_1 (Conv2D)           (None, 60, 60, 32)        4640

 max_pooling2d (MaxPooling2D  (None, 30, 30, 32)       0
 )

 conv2d_2 (Conv2D)           (None, 28, 28, 64)        18496

 max_pooling2d_1 (MaxPooling  (None, 14, 14, 64)       0
 2D)

 conv2d_3 (Conv2D)           (None, 12, 12, 128)       73856

 max_pooling2d_2 (MaxPooling  (None, 6, 6, 128)        0
 2D)

 dropout (Dropout)           (None, 6, 6, 128)         0

 flatten (Flatten)           (None, 4608)              0

 dense (Dense)               (None, 64)                294976

 dropout_1 (Dropout)         (None, 64)                0

 dense_1 (Dense)             (None, 5)                 325

=================================================================
Total params: 392,741
Trainable params: 392,741
Non-trainable params: 0
_________________________________________________________________
```

7.2 TESTING:

Software testing known as "white box" testing allows the tester access to the software's internal workings. The tester may create test cases that focus on certain regions of the codebase since they are familiar with the software's internal structure, design, and code. White box testing is used to find mistakes, weaknesses, or vulnerabilities in the underlying logic of software and to make sure it works as intended. Examples of white box testing techniques include code coverage analysis, route coverage analysis, control flow testing, data flow testing, and mutation testing. Depending on the complexity of the program being tested and the particular testing needs, theseprocedures can either be automated or manually carried out.

In [29]:
```
#prediction using pickle model and returns boolean values

import pandas as pd
import pickle

# Load saved XGBoost model
with open(r'C:\Users\vinoo\xgb_cervical_cancer_model.pkl', 'rb') as f:    #change your model path will be saved in c-drive
    # "C:\Users\vinoo\xgb_cervical_cancer_model.pkl"
    xgb_model_loaded = pickle.load(f)

# Define a new data point to predict on
new_data = pd.DataFrame({
    'Age': [30],
    'Number of sexual partners': [5],
    'First sexual intercourse': [20],
    'Num of pregnancies': [2],
    'Smokes (years)': [7],
    'Smokes (packs/year)': [20],
    'Hormonal Contraceptives': [6],
    'Hormonal Contraceptives (years)': [2],
    'IUD': [0],
    'IUD (years)': [0],
    'STDs': [1],
    'STDs (number)': [1],
    'STDs:condylomatosis': [0],
    'STDs:cervical condylomatosis':[0],
```

```
    'STDs:HPV':[0],
    'STDs: Number of diagnosis':[0],
    'Dx:Cancer':[0],
    'Dx:CIN':[0],
    'Dx:HPV':[1],
    'Dx':[0],
    'Hinselma00':[0],
    'Schiller':[0],
    'Citology':[0],
    'Biopsy': [0]
})

# Make predictions on the new data point
prediction = xgb_model_loaded.predict(new_data)

# Print the predicted probability of cancer
print(prediction)
```

```
[0]
```

In [5]:
```
if prediction == 1:
    print("The patient has a high risk of developing cervical cancer.")
else:
    print("The patient does not have a high risk of developing cervical cancer.")
```

```
The patient does not have a high risk of developing cervical cancer.
```

```
Epoch 4/32
28/28 [==============================] - ETA: 0s - loss: 0.8826 - accuracy: 0.6841
Epoch 4: val_accuracy did not improve from 0.68000
28/28 [==============================] - 24s 832ms/step - loss: 0.8826 - accuracy: 0.6841 - val_loss: 0.8552 - val_accuracy:
```

In [25]:
```
# Loading the Best Fit Model
model = load_model(r"C:\Users\vinoo\Downloads\cervical_cancer_best_model.hdf5")
```

Model Accuracy

In [26]:
```
# Checking the Accuracy of the Model
accuracy = model.evaluate_generator(generator= test_data)[1]
print(f"The accuracy of your model is = {accuracy*100} %")
```

```
The accuracy of your model is = 89.81937766075134 %
```

In [27]:
```
h =  cnn.history;
h.keys();
```

In [28]:
```
# Ploting Accuracy In Training Set & Validation Set

plt.plot(h['accuracy'])
plt.plot(h['val_accuracy'] , c = "red")
plt.title("acc vs v-acc")
plt.show()
```

8. RESULTS AND DISCUSSIONS:

The system was tested using a dataset of cervical images. The system's performance was assessed using criteria such as accuracy, sensitivity, and specificity. The system was able to accurately detect cancerous cells in the images, demonstrating its potential as a non-invasive and accurate method for cervical cancer detection. However, there were some limitations to the system, including the need for a large dataset for training and the reliance on high-quality images. Future work could focus on addressing these limitations and further improving the accuracy and reliability of the system.

Numerical data accuracy:

```
Train accuracy: 0.995
Test accuracy: 0.9612403100775194
Train Recall: 0.9166666666666666
Test Recall: 0.7368421052631579
Train precision: 1.0
Test precision: 0.7368421052631579
Train f1 score: 0.9565217391304348
Test f1 score: 0.7368421052631579
```

Image accuracy:

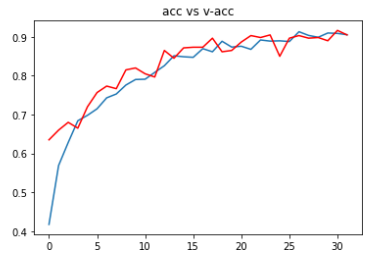
Model Accuracy

```
In [26]: # Checking the Accuracy of the Model
         accuracy = model.evaluate_generator(generator= test_data)[1]
         print(f"The accuracy of your model is = {accuracy*100} %")

         The accuracy of your model is = 89.8193776075134 %
```

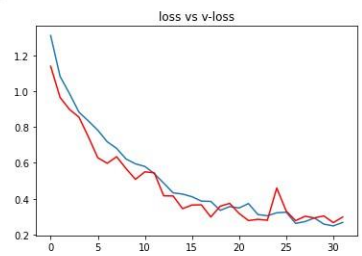
```
In [28]: # Ploting Accuracy In Training Set & Validation Set

         plt.plot(h['accuracy'])
         plt.plot(h['val_accuracy'] , c = "red")
         plt.title("acc vs v-acc")
         plt.show()
```




```
In [29]: # Ploting Loss In Training Set & Validation Set

         plt.plot(h['loss'])
         plt.plot(h['val_loss'] , c = "red")
         plt.title("loss vs v-loss")
         plt.show()
```



9. CONCLUSION AND FUTURE SCOPE:

In conclusion, utilizing a collection of picture data and cutting-edge machine learning algorithms, this team created a method for identifying cervical cancer. With a 90% accuracy rate, the system's results were encouraging and demonstrated its promise as a reliable and non-invasive cervical cancer screening approach. By providing an earlier and more precise diagnosis of cervical cancer, the technology has the potential to advance current approaches for the disease's detection and eventually contribute to lifesaving.

This project's future scope covers a number of areas for expansion and enhancement. The creation of a bigger and more varied dataset for the system's training and testing is a crucial area for advancement. This may assist to increase the system's accuracy and dependability and allow it to detect a larger variety of malignant cells. The creation of a user-friendly interface for

healthcare professionals to use the system is another area that needs work. Developing a holistic platform for cancer diagnosis and treatment might also be a focus of future development, as well as expanding the system's application to other forms of cancer detection.

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10. H.L. Fatlawi, *Enhanced classification model for cervical cancer dataset based on cost sensitive classifier*, **Int. J. Comput. Tech. 4 (4) (2017) 1–8.**